

GCE A LEVEL
A520U10-1 (Predicted – Advanced)
ECONOMICS – A level Component 1
Economic Principles

1 hour 30 minutes

ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

You are reminded of the necessity for good English and orderly presentation in your answers.

For Examiner's use only

Question	Maximum Mark	Mark Awarded
1–20	20	
21.	4	
22.	7	
23.	4	
24.	8	
25.	8	
26.	9	
Total	60	

Answer all questions.

SECTION A

For each question in Section A, write the letter (A, B, C, D or E) that corresponds to your answer in the box provided.

You are advised to spend approximately 30 minutes on this section.

1. A firm's total cost function is $TC = 200 + 30Q - 2Q^2 + 0.2Q^3$. At output $Q = 10$, the average fixed cost and marginal cost are: **[1]**

- A** AFC = £20, MC = £20
- B** AFC = £20, MC = £50
- C** AFC = £30, MC = £50
- D** AFC = £30, MC = £20
- E** AFC = £50, MC = £30

Answer: _____

2. In a Cournot duopoly with homogeneous products, linear inverse demand $P = 120 - Q$, and each firm facing a constant marginal cost of £30, the symmetric Nash equilibrium output for each firm is: **[1]**

- A** 22.5
- B** 30
- C** 45
- D** 60
- E** 90

Answer: _____

3. An economy has a marginal propensity to consume of 0.8, a marginal tax rate of 0.25, and a marginal propensity to import of 0.15. The Keynesian multiplier is approximately: **[1]**

- A** 1.67
- B** 1.82
- C** 2.00
- D** 2.86
- E** 5.00

Answer: _____

4. The yield curve inverts (short-term interest rates exceed long-term rates). Which of the following is the most common interpretation by financial markets? **[1]**

- A** Investors expect short-term interest rates to rise further
- B** Investors expect a recession, during which the central bank will cut rates
- C** Inflation expectations have risen sharply
- D** The government plans to issue more long-dated debt
- E** Investors have become more risk-averse on short-duration bonds

Answer: _____

5. A price-discriminating monopolist faces two separated markets with demand $P_1 = 100 - Q_1$ in market 1 and $P_2 = 80 - 2Q_2$ in market 2. MC is constant at £20. The profit-maximising prices are: **[1]**

- A** $P_1 = £40$, $P_2 = £30$
- B** $P_1 = £50$, $P_2 = £40$
- C** $P_1 = £60$, $P_2 = £50$

D $P_1 = £40$, $P_2 = £50$

E $P_1 = £70$, $P_2 = £60$

Answer: _____

6. A contestable market is characterised by: **[1]**

A Many firms producing homogeneous products with perfect information

B Low barriers to entry and exit, meaning that the threat of entry disciplines incumbent behaviour even if the actual number of firms is small

C A single firm earning supernormal profit protected by regulation

D Firms that collude using tacit agreements to fix prices

E Firms that engage in both first- and third-degree price discrimination

Answer: _____

7. The chart shows US Treasury yield data: **[1]**

Bond maturity	2022 yield (%)	2024 yield (%)
3-month	0.08	5.35
2-year	0.73	4.85
10-year	1.52	4.30
30-year	1.90	4.50

Which of the following conclusions is best supported by the data?

A The yield curve was inverted in 2022 and upward-sloping in 2024

B The yield curve was upward-sloping in 2022 and has partially inverted in 2024

C Inflation expectations were higher in 2022 than in 2024

D Long-term bond prices have risen between 2022 and 2024

E Risk premiums have fallen for all maturities

Answer: _____

8. Okun's law, in its typical formulation, implies that: **[1]**

A A 1% rise in inflation is associated with a 1% fall in unemployment

B A 1 percentage point rise in unemployment is associated with roughly a 2% fall in real GDP relative to trend

C Unemployment and inflation always move in the same direction

D Productivity growth determines the unemployment rate in the long run

E Labour force participation is unrelated to the business cycle

Answer: _____

9. A government imposes a tradable pollution permit scheme with a fixed number of permits. Compared to a Pigouvian (emissions) tax at the same initial marginal abatement cost, the permit system: **[1]**

A Guarantees a particular emissions quantity but allows the price of permits to vary with demand

B Guarantees a particular permit price but allows emissions quantity to vary

C Always raises more government revenue

D Is equivalent under all conditions

E Cannot lead to the social optimum

Answer: _____

10. The 'Dutch disease' hypothesis describes a situation in which: [1]

- A Currency depreciation damages exporters
- B A boom in one tradable sector (often resource extraction) causes real exchange rate appreciation that damages other tradable sectors
- C Manufacturing declines due to automation
- D Tulip markets overheat and collapse
- E Rising productivity always causes unemployment

Answer: _____

11. In a Solow growth model, a country in steady state with technology growth rate g , population growth rate n , and depreciation rate δ will have real output per worker growing at a rate of: [1]

- A 0
- B n
- C g
- D $n + g$
- E $n + g + \delta$

Answer: _____

12. Quantitative easing (QE) by the central bank would, in theory, most directly: [1]

- A Increase the monetary base by creating central bank reserves to purchase long-dated assets
- B Reduce the money multiplier by raising reserve requirements
- C Raise the policy interest rate
- D Reduce the stock of government debt outstanding in private hands and cannot affect long bond yields
- E Increase the fiscal deficit

Answer: _____

13. A monopsonist in a labour market has marginal cost of labour MCL that lies above the labour supply (wage) curve because: [1]

- A Trade unions impose a wage floor
- B Monopsonists are forced to pay all existing workers the higher wage required to attract an additional worker
- C Workers have monopoly power
- D The supply curve of labour is perfectly elastic
- E Marginal product of labour diminishes

Answer: _____

14. The chart shows UK labour productivity (output per hour, 2015 = 100): [1]

Year	Output per hour
2015	100.0
2018	102.3
2021	103.1
2022	101.8
2023	102.0

2024	101.5
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Which of the following is the most appropriate description of UK productivity over this period?

- A Productivity has grown at approximately the long-term trend rate of 2% per year
- B Productivity has stagnated at far below the historical trend rate
- C Productivity growth has accelerated sharply
- D Productivity has fallen every year since 2015
- E Productivity has matched the G7 average

Answer: _____

15. An economy with a rigid real wage above the equilibrium level will experience: [1]

- A Frictional unemployment only
- B Cyclical unemployment
- C Classical (real-wage) unemployment, because labour supply exceeds labour demand at the prevailing wage
- D Disguised unemployment
- E No unemployment if productivity is high enough

Answer: _____

16. The Heckscher–Ohlin theorem predicts that a country will export: [1]

- A Goods in which it has the highest labour productivity
- B Goods that intensively use its relatively abundant factor of production
- C Goods that have the highest value-added
- D Goods it produces using the most advanced technology
- E Goods protected by tariffs abroad

Answer: _____

17. The Lucas critique argues that: [1]

- A Macroeconomic data are too noisy to be useful
- B Estimated parameters from historical data may not be valid when policy changes, because economic agents will adjust their expectations and behaviour
- C Fiscal policy is always ineffective
- D Rational expectations cannot be formed
- E Governments should target money supply instead of inflation

Answer: _____

18. A country runs a persistent current account deficit equal to 5% of GDP financed mainly by portfolio inflows. Which of the following would represent the greatest macroeconomic risk? [1]

- A A rise in domestic saving
- B A sudden stop – foreign investors lose confidence and rapidly withdraw capital
- C An improvement in the terms of trade
- D A rise in domestic export productivity
- E A depreciation of the currency

Answer: _____

19. A monopolist with demand $P = 200 - 2Q$ and cost function $TC = 50 + 20Q + Q^2$ will earn economic profit of: [1]

- A £2,000
- B £2,350

- C** £2,650
- D** £2,900
- E** £3,200

Answer: _____

20. Which of the following is most consistent with 'secular stagnation' as articulated by Larry Summers? **[1]**

- A** Temporarily weak demand during a recession
- B** A sustained imbalance in which desired saving exceeds desired investment at a zero real interest rate, producing chronically weak demand and low growth
- C** High inflation caused by monetary loosening
- D** A one-off productivity slowdown after a financial crisis
- E** A rise in the natural rate of unemployment

Answer: _____

SECTION B

21. The charts below show (1) the US 10-year minus 2-year Treasury yield spread (%) and (2) the year-on-year change in US real GDP (%).

Chart 1 – US 10Y–2Y yield spread (%)

2019: +0.21 | 2020: +0.54 | 2021: +1.12 | 2022: –0.48 | 2023: –0.42 | 2024: –0.28

Chart 2 – US real GDP year-on-year growth (%)

2019: +2.3 | 2020: –2.2 | 2021: +5.8 | 2022: +2.1 | 2023: +2.5 | 2024: +2.8

Using the data, explain the relationship between the two charts and comment on its reliability as a recession indicator. **[4]**

22. The diagram below illustrates the market for a good with negative production externalities, alongside a marginal abatement cost (MAC) curve for pollution associated with production.

[Diagram 1: a standard market diagram with MPC below MSC, $D = MSB$, free-market equilibrium Q_m at $MPC = MSB$ above social optimum Q_s at $MSC = MSB$. Welfare loss triangle shown between MSC and MSB from Q_s to Q_m .]

[Diagram 2: a MAC curve (downward sloping in emissions reduced) showing that the first units of abatement are cheap but marginal abatement cost rises rapidly at high abatement levels. The social optimum is where $MAC =$ marginal external damage.]

Using the diagrams, discuss whether a tradable pollution permit system is superior to a Pigouvian (emissions) tax when there is uncertainty about the slope of the marginal abatement cost curve. **[7]**

23. An incumbent monopolist (Firm I) has invested in excess production capacity. A potential entrant (Firm E) is deciding whether to enter the market. If Firm E enters, Firm I chooses whether to Fight (use its excess capacity to flood the market) or Accommodate (share the market). The payoff matrix (Firm E's profit, Firm I's profit, £m) is:

		Firm I (response if Firm E enters)	
		Fight	Accommodate
Firm E	Enter	–£20m, £30m	£25m, £60m
	Stay out	£0, £120m	£0, £120m

Use the data to explain the Nash equilibrium and whether Firm I's threat to fight would be credible.
[4]

24. The chart and extract below describe UK productivity.

Chart: UK output per hour (index, 2015 = 100)

2015: 100.0 | 2017: 100.9 | 2019: 101.8 | 2020: 103.4 | 2021: 103.1 | 2022: 101.8 | 2023: 102.0 | 2024: 101.5

Comparison: G7 average output per hour grew by approximately 8% over the 2015–2024 period.

Extract:

The UK's 'productivity puzzle' has persisted since the 2008 financial crisis. Before 2008, UK output per hour grew by around 2% annually; since then, the average has been under 0.5% per year. Weak business investment, low R&D spending (1.7% of GDP vs OECD average 2.6%), skills shortages, regional imbalances and a 'long tail' of unproductive firms are frequently cited as causes.

The government is considering a package of supply-side reforms including: (i) permanent full expensing of capital investment; (ii) expanded technical training and apprenticeships; (iii) planning reform to accelerate infrastructure delivery; (iv) enhanced R&D tax credits; and (v) measures to increase labour force participation.

Using the data, discuss whether such supply-side policies are likely to raise the UK's long-run growth rate significantly. **[8]**

25. The table below provides data on four countries.

	HDI	GNI/capita (PPP \$)	Gini coefficient	IHDI (inequality- adjusted)	MPI (multidimensi onal poverty)
South Africa	0.717	13,468	0.630	0.478	0.025
Sri Lanka	0.782	13,086	0.377	0.697	0.011

Note: The Gini coefficient ranges from 0 (perfect equality) to 1 (perfect inequality). The MPI (multidimensional poverty index) measures acute poverty across health, education and living standards; lower values are better.

Discuss whether, on the basis of the data provided, Sri Lanka can be considered more developed than South Africa. **[8]**

26. The chart and extract below give information on the UK housing market.

Chart: UK average house price (£000) and 2-year fixed mortgage rate (%)

2020 Q1: £235k, 1.8% | 2021 Q1: £251k, 1.6% | 2022 Q1: £277k, 1.6% | 2023 Q1: £290k, 4.4% |
2024 Q1: £282k, 5.8%

Extract:

UK house prices surged during the pandemic, rising by nearly 25% between 2020 and 2022, driven by ultra-low interest rates, stamp duty holidays, and demand for larger homes following the shift to remote working. Prices have since softened as Bank Rate rose from 0.1% to 5.25% and mortgage rates reached 15-year highs.

On the supply side, the UK has consistently built fewer homes than its estimated housing need of 300,000 per year – only around 235,000 were built in 2023. Restrictive planning rules, a shortage of construction workers, and high material costs constrain new supply. Meanwhile, strong net migration and the formation of smaller household units (e.g. more single-person households) continues to support underlying demand.

Looking forward, the Bank of England has signalled that interest rates may have peaked, and the government is exploring further planning reforms to boost supply.

Using supply and demand diagram(s), discuss whether UK house prices are more likely to rise or fall over the next few years. **[9]**

END OF PAPER